

Introduction/Key Drivers in the Design Process

1.1 Introduction

The primary area of emphasis in the text is the methods and technology for aerodynamic configuration sizing (Fig. 1.1). Aerodynamic configuration sizing is conducted to develop the configuration geometry and the dimensions of the missile. The output of aerodynamic configuration sizing includes the missile diameter, length, nose geometry, stabilizer size and geometry, and the control surface size and geometry. The configuration sizing methods of this text are physics-based methods to allow analytical sizing of the missile configuration. Areas that also receive primary emphasis are aerodynamic stability and control, aerodynamic flight performance, propulsion, structure, and weight. Other areas that receive secondary emphasis are warhead, miss distance, cost, additional measures of merit, and launch platform integration. These are addressed to a level that is consistent with their impact on the aerodynamic configuration sizing. Tertiary emphasis is given to the areas of seeker, sensors, electronics, and power supply. The areas of safe, arm, and fuzing are not addressed in this text. Although these areas are important at the system level, they generally have lesser impact on the aerodynamic design.

1.2 Tactical Missile Characteristics

Tactical missiles are different from other flight vehicles, such as combat aircraft, and are a technical specialty in their own right. Figure 1.2 is a state-of-the-art comparison of tactical missile characteristics with the current state of the art (SOTA) of fighter aircraft. Examples are shown where missiles are driving technology. Also shown are other areas where the missile is not driving technology.

As an example, the lateral and longitudinal acceleration SOTA of missiles exceeds that of combat aircraft. Missile lateral maneuverability of $30g^+$ and longitudinal acceleration of $30g^+$ have been demonstrated. Notable examples of missiles with high acceleration and maneuverability include the AGM-88 HARM and the AA-11 Archer missiles. Developmental missiles have demonstrated lateral maneuverability of $100g^+$ and longitudinal acceleration of $400g^+$. Missile speed and altitude may be greater than that of combat aircraft. An example of a high-speed endo-atmospheric missile is the MM-3 hypervelocity

Area	Emphasis
Aero Configuration Sizing	●
Aero Stability & Control	●
Aero Flight Performance	●
Propulsion	●
Structure	●
Weight	●
Warhead	◐
Miss Distance	◐
Cost	◐
Additional Measures of Merit	◐
Launch Platform Integration	◐
Seeker, Sensors, and Electronics	○
Power Supply	○
Safe, Arm, and Fuzing	○

Primary Emphasis

●

Secondary Emphasis

◐

Tertiary Emphasis

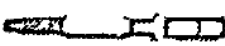
○

Not Addressed

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Fig. 1.1 Emphasis of this text is on physics-based, analytical sizing of the aerodynamic configuration.

missile. An example of a high-speed exo-atmospheric missile is the SM-3 standard missile. Another difference is the dynamic pressure loading on a missile, which is usually greater than that of combat aircraft. For example, the PAC-3 missile operates at much higher dynamic pressure than that of aircraft. Another

Tactical Missile Characteristics	Example of State of the Art	Comparison With Fighter Aircraft
Axial Acceleration	 AGM-88	●
Maneuverability	 AA-11	●
Speed / altitude	 SM-3	●
Dynamic pressure	 PAC-3	●
Size	 Javelin	●
Weight	 FIM-92	●
Production cost	 GBU-31	●
Observables	 AGM-129	○
Range	 AGM-86	-
Kills per use	 Storm Shadow	-
Target acquisition	 LOCAAS	-

● Superior

◐ Better

○ Comparable

- Inferior

Fig. 1.2 Tactical missiles are different from fighter aircraft.

difference is the relatively small size and lighter weight of missiles in comparison to combat aircraft, examples including the Stinger and Javelin man-portable missiles. Related to cost, missiles are a throwaway. As a result, they are more cost driven than combat aircraft. Development cost is lower for missiles, and the difference in production cost is even more dramatic. An example is the GBU-31 JDAM, with unit cost on the order of \$10,000 to \$20,000, compared to a cost of tens of millions of dollars for typical combat aircraft.

Cruise missiles such as AGM-129 can have radar cross section (RCS) comparable to that of low observable aircraft. A cruise missile can achieve low RCS without the same limitations for aircraft such as pilot integration and cockpit, windshield, and fire control sensors. However, a design limitation for cruise missiles is launch platform integration, which may limit the amount of low RCS configuration shaping and tailoring that is practical with a span constraint.

Areas where the combat aircraft generally have superior capability include range, targets killed per use, and target acquisition. Although the conventional version of the AGM-86 cruise missile (CALCM) has a flight range that can exceed 700 n miles, combat aircraft can have a much longer range. In the area of target kill capability, precision strike missiles have become more efficient in recent years, with a single target kill probability approaching one and a capability for multiple target kills. The Storm Shadow and Scalp are examples of efficient precision strike missiles. Precision strike missiles have high accuracy and are capable of dispensing submunitions, exhibiting high firepower. However, the same missiles are load-outs on combat aircraft, and so the enhancement in missiles also enhances the combat aircraft effectiveness and firepower. Finally, although missiles such as Brimstone and smart powered submunitions such as the Low Cost Autonomous Attack Submunition (LOCAAS) have demonstrated a capability for automatic target recognition (ATR), a combat aircraft with a human pilot continues to have superior capability for target recognition, discrimination, and acquisition. Autonomous target acquisition by missiles is a relatively immature technology that will improve in the future with new technologies such as multimode and multispectral seekers. A lower risk, but more human logistical intensive alternative to autonomous target acquisition is to use real-time two-way data links.

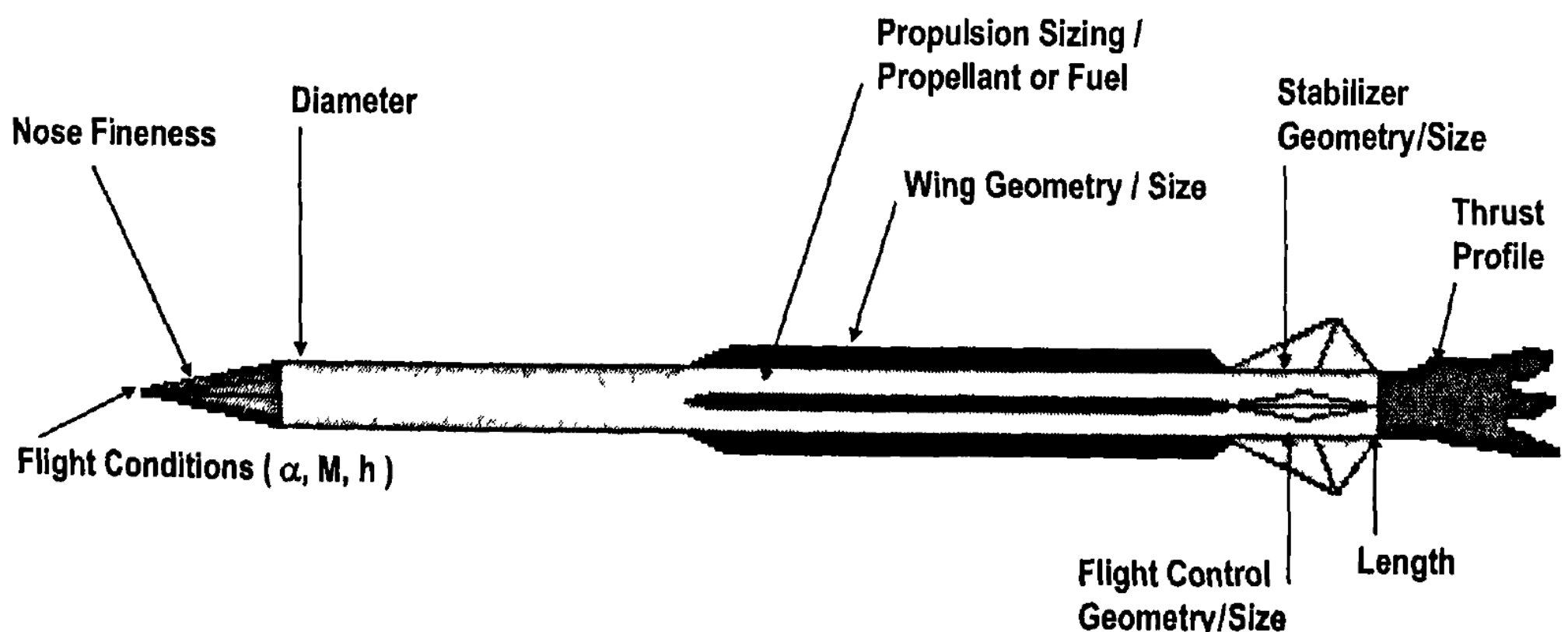
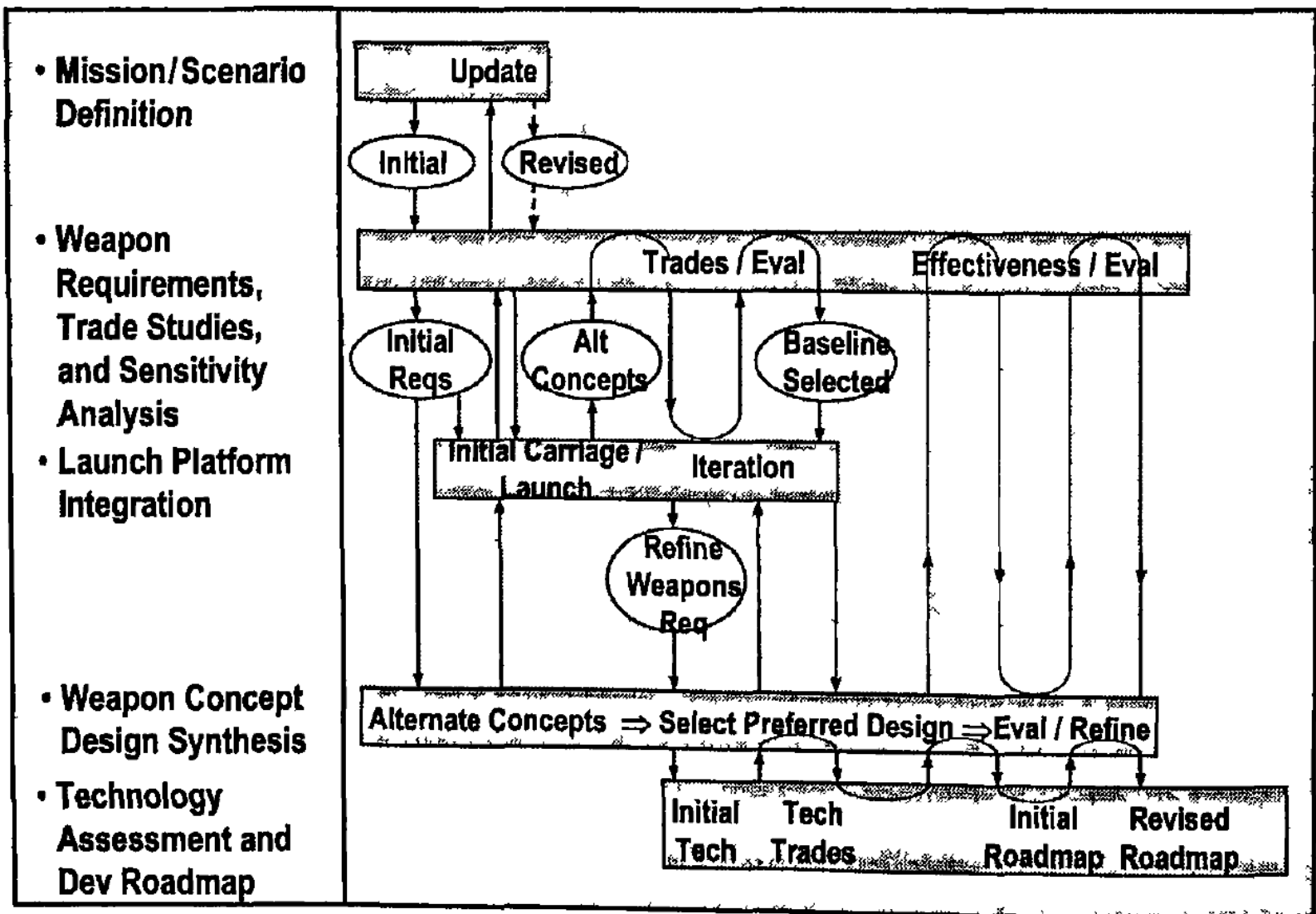


Fig. 1.3 Aerodynamic configuration sizing parameters emphasized in this text.

Figure 1.3 is a summary of the aerodynamic configuration sizing parameters emphasized in this text. These are flight conditions (α , M , h), nose fineness, diameter, propellant/fuel type and weight, wing geometry/size, stabilizer geometry/size, flight control geometry/size, length, and thrust profile. Flight condition parameters that are most important in the design of tactical missiles are angle of attack α , Mach number M , and altitude h . For the aerodynamic configuration, the missile diameter and length have a first-order effect on characteristics such as missile drag, subsystem packaging available volume, launch platform integration, seeker and warhead effectiveness, and body bending. Another configuration driver is nose fineness, an important contributor to the drag of supersonic missiles. Nose fineness also affects seeker performance, available propellant length, and missile observables. Another example is missile propellant/fuel type and weight, which drive flight performance range and velocity. The aerodynamic configuration wing geometry and size are often set by maneuverability requirements and aerodynamic efficiency. Stabilizer geometry and size are often established by static margin requirements. In the flight control area, the geometry and size of the flight control surfaces determine the maximum achievable angle of attack and the resulting maneuverability. Finally, the thrust profile determines the missile velocity time history.

1.3 Conceptual Design Process

Figure 1.4 shows a typical conceptual design process for missiles. Conceptual design is an iterative process, requiring a number of design iterations to achieve a balance of emphasis from the diverse inputs and outputs. The major tasks of conceptual design are 1) mission/scenario definition; 2) weapon requirements, trade



Note: Typical conceptual design cycle is 3 to 9 months. House of quality may be used to translate customer requirements to engineering characteristics. DOE may be used to efficiently evaluate the broad range of design solutions.

Fig. 1.4 Conceptual design process requires evaluation of alternatives and iteration.

studies, and sensitivity analysis; 3) physical integration of the missile with the launch platform; 4) weapon concept design synthesis; and 5) technology assessment and technology development roadmap. The initial design process begins with a general definition of the mission/scenario. The input is a “requirements pull” of the desired capability from the military customer. The customer input for mission requirements is usually not changed during the design study. However, sometimes the mission requirements may be changed by the customer during the design study. Examples are if they are found to be too demanding, too expensive, or the “technology push” of potential technology availability that is provided by the technical community shows that a new technology can be better utilized by different mission requirements. A house of quality may be used to relate the customer requirements and their relative weighting to the engineering design characteristics. The weapon requirements, trade studies, and sensitivity analysis task provide the high level requirements on the missile such as range, time to target, and other measures of merit. This task is oriented toward an operations analysis of a system-of-systems, including command, control, and communication (C³) targeting. The high level requirements that were initially provided by the military customer are refined through computer modeling. The system-level computer models may be characterized as campaign, raid, or engagement models. In a campaign model many different types of systems are interacting over a simulated time interval from days to weeks. A raid model has multiple platforms engaging multiple targets during a single raid. An engagement model may be that of a single launch platform and missile engaging a single target or threat. The third task, physical integration of the missile with the launch platform, provides constraints such as the length, span, and weight. This task is oriented toward systems integration. The fourth task, weapon concept design synthesis, is the most iterative and arguably the most creative. Characteristics such as the aerodynamic shape, propellant or fuel type and weight, flight trajectory range, time to intercept, maneuverability, seeker detection range, accuracy, lethality, and cost are evaluated; and the missile is resized and reconfigured in an iterative process. For example, the tail stabilizers and flight control surfaces may be resized for improved stability or maneuverability. Another example is adding propellant or fuel to match the flight range requirement. As the design matures and becomes better defined through iteration, the number of alternative solutions is reduced from a broad range of possibilities to a smaller set of preferred candidates that are more reasonable and cost effective. More in-depth information is provided for the design subsystems as the design matures. Finally, a technology assessment task further defines the subsystems and selects the best technologies from the candidate approaches. The technology trades lead to a set of preferred, enabling technologies. A technology roadmap documents the development plan for maturing the enabling technologies.

A typical duration for a conceptual design activity is three to nine months. The products of the missile design activity include refined mission/scenario definitions, system-of-systems definition of the missile requirements, and the interface of the missile with other system elements such as targeting, launch platform compatibility compliance, advanced missile concepts, identification of the enabling technologies, and a technology roadmap.

Figure 1.5 shows the evaluation of alternatives and the iterative process used for conceptual design synthesis. Based on mission requirements, an initial baseline from an existing missile with similar propulsion is established. It is used as a starting point to expedite the design convergence. Advantages of a baseline missile include the prior consideration of balanced system engineering for the subsystems and the use of an accurate benchmark based on existing test data (e.g., wind-tunnel data). Changes are made in the baseline missile aerodynamics, propulsion, weight, and flight trajectory to reflect the new requirements of the new missile concept. The new conceptual design is evaluated against its flight performance requirements (e.g., range, time to target, maneuver footprint). The aerodynamics portion of the conceptual design process is an investigation of alternatives in configuration geometry. The output of the aerodynamics calculation is then inputted to the propulsion system design to size the propulsion system. Propulsion sizing includes providing sufficient propellant or fuel to meet the range and time-to-target requirements. The next step is to estimate the weight of the new missile with its modified aerodynamics and propulsion. Much of this activity is focused on structural design, which is sensitive to changes in flight performance. Following the weight sizing, flight trajectories are computed for the new missile. The range, terminal velocity, maneuverability, and other flight performance parameters are then compared with the mission flight performance requirements. If the missile does not meet the flight performance requirements, it is resized and reiterated. After completing a sufficient number of iterations to meet the flight performance requirements, the next step

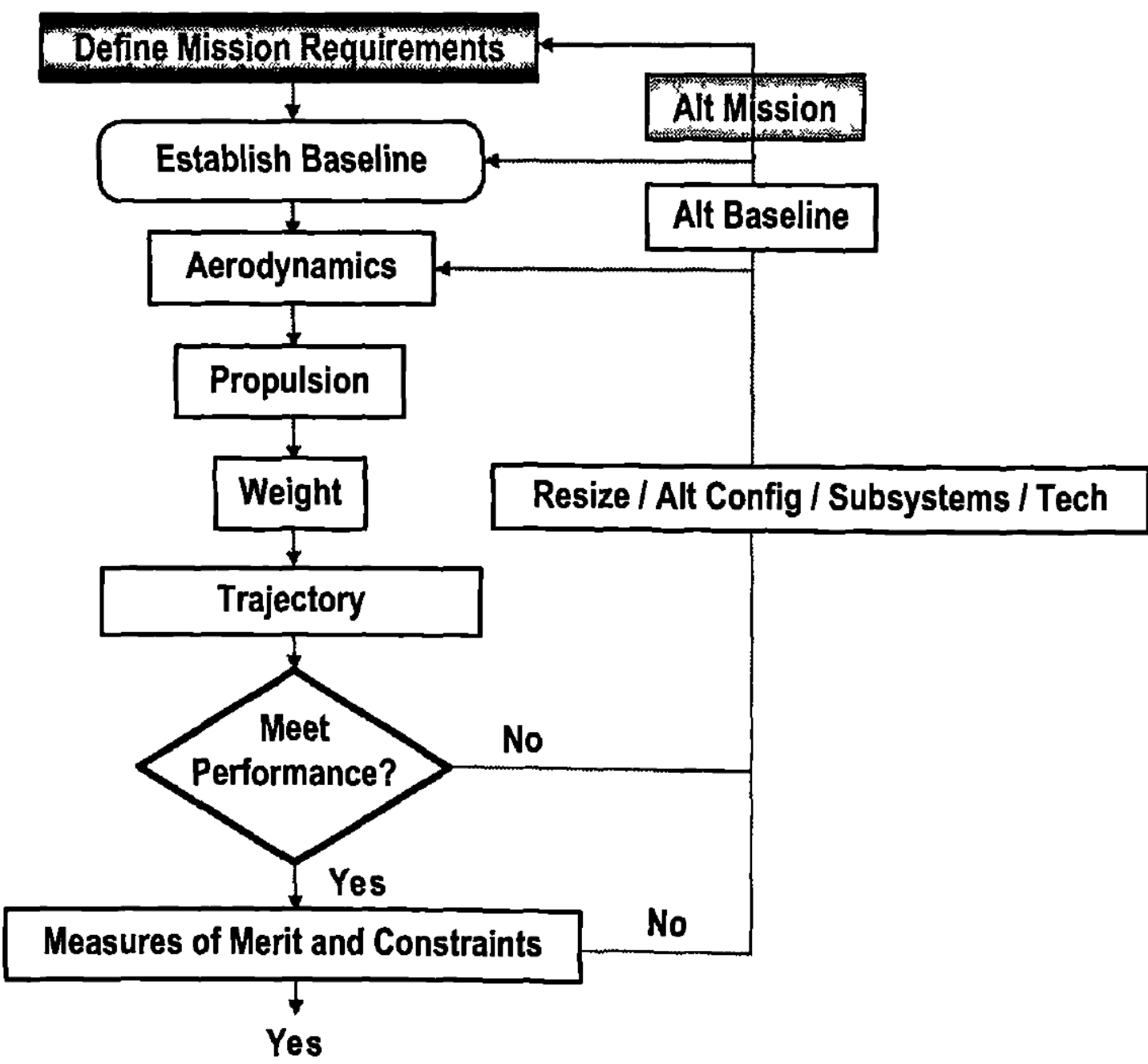


Fig. 1.5 Missile concept synthesis requires evaluation of alternatives and iteration.

is evaluating the new missile against the other measures of merit and constraint requirements. If the missile does not meet the requirements, the design is changed (alternative configuration, subsystems, technologies) and resized for the next iteration and evaluation.

A synthesized missile will differ from the starting point baseline in several respects. For example, the wing area may have been resized to meet the maneuverability requirement. The tail area may have been resized to meet static margin and maximum trim angle-of-attack requirements. The rocket motor or the ramjet engine may have been modified to improve its efficiency at the selected design altitude or Mach number. Additionally, the length of the propulsion system may have been changed to accommodate additional propellant/fuel necessary to satisfy flight range requirements. The design changes are reflected in revisions to the mass properties, configuration geometry, thrust profile, and flight trajectory for the missile. Typically, three to six design iterations are required before a synthesized missile converges to meet the flight performance requirements.

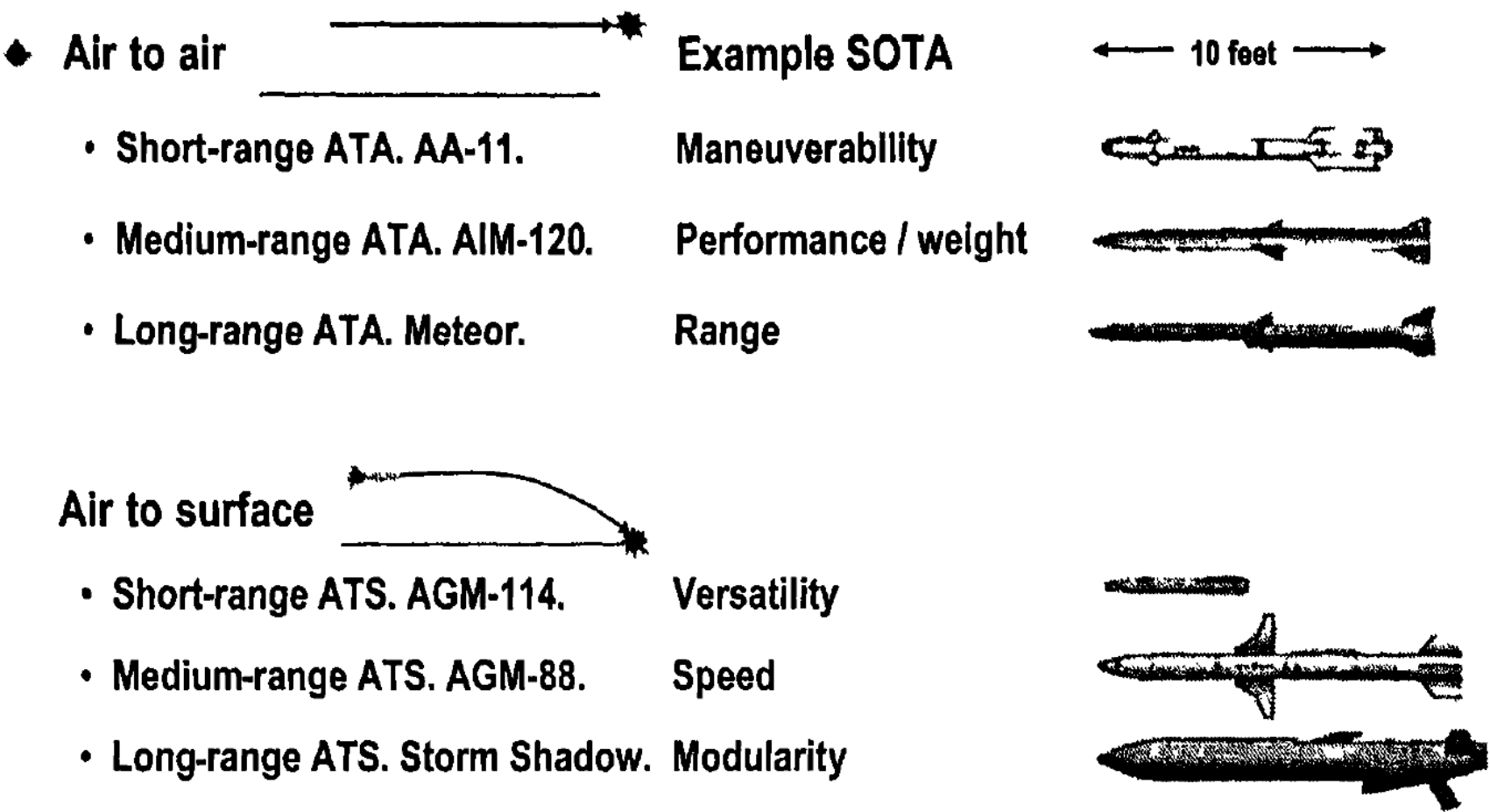
Although a missile design may meet performance requirements and satisfy the other measures of merit and launch platform constraints, it is not necessarily an optimum missile design. It is good design practice to evaluate as many design alternative concepts as possible within the time duration of the design study. A design of experiments (DOE) is an efficient approach to evaluate the broad set of alternatives.

Following conceptual design, the missile is further refined in preliminary design. Preliminary design activities include parameter optimization and ground tests.

1.4 Examples of State-of-the-Art Missiles

Shown in Figs. 1.6 and 1.7 are examples of state-of-the-art missiles for the mission areas of air to air, air to surface, surface to surface, and surface to air. Reference 1 was a source for the missile drawings. Each mission area has examples of short-, medium-, and long-range missiles. Short-range missiles operate within visual range engagements, less than about 3 n miles, have rocket propulsion, and are usually guided by an electro-optical (e.g., infrared, laser) seeker, with seeker lock-on before launch. Medium-range missiles can operate beyond visual range engagements out to the earth's horizon, about 20 n miles, have rocket propulsion, and are usually guided by a radar seeker. Most medium-range missiles also have an inertial reference system and are capable of seeker lock-on after launch. Long-range missiles have ranges greater than the earth's horizon. Long-range missiles have combined inertial/seeker guidance and are capable of seeker lock-on after an inertial midcourse fly-out to the vicinity of the target. Long-range tactical missiles may have rocket, ducted rocket, ramjet, or turbojet propulsion, depending upon their speed and range requirements. Tactical ballistic missiles are rocket powered, while cruise missiles have air-breathing propulsion.

Three examples are shown in the top of Fig. 1.6 of the state-of-the-art drivers of air-to-air missiles. The selected drivers are air-to-air maneuverability, performance with light weight, and range. An example of a short-range air-to-air missile is the highly maneuverable AA-11 Archer. An example of a medium-range missile

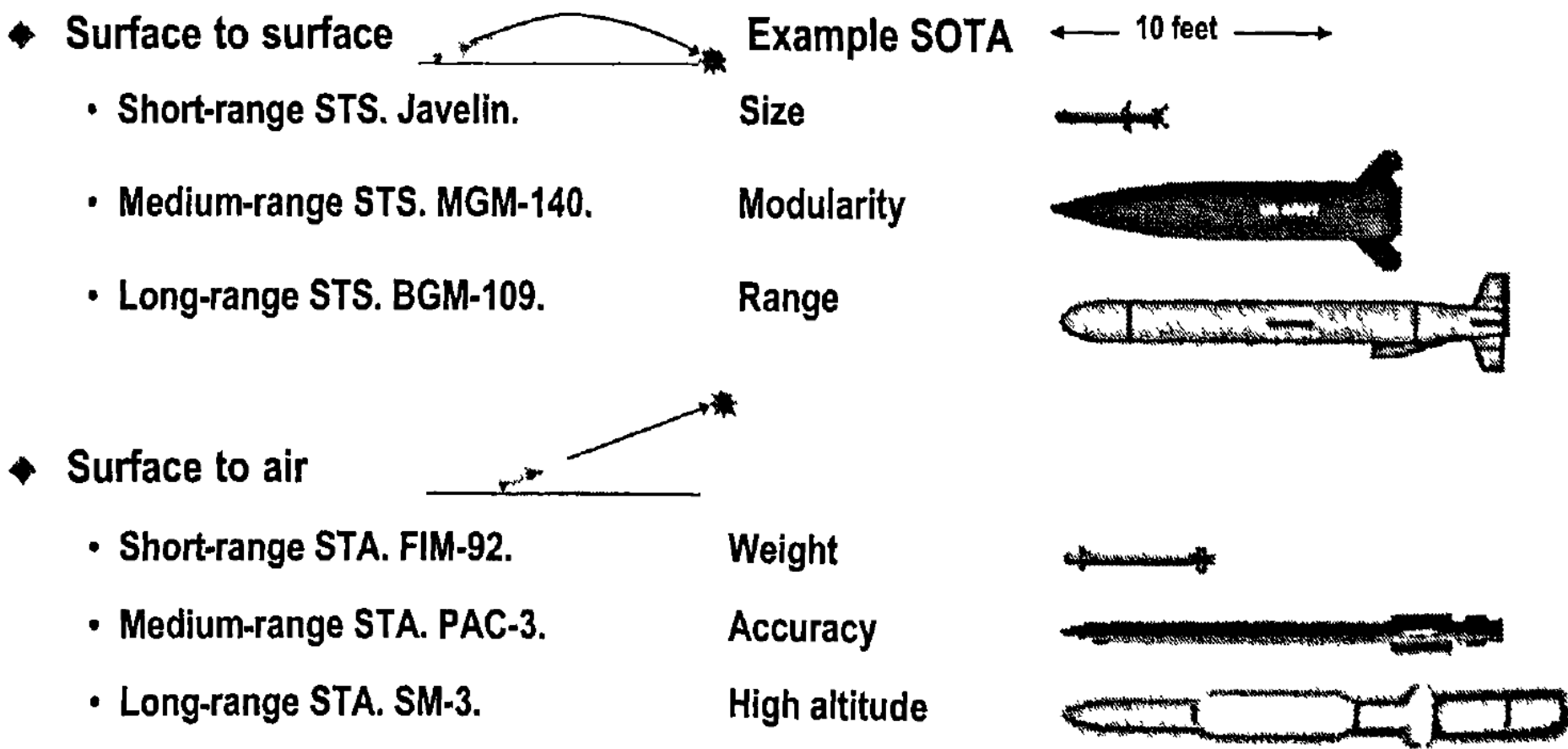


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Fig. 1.6 Examples of air-launched missile missions/types.

is the high-performance/lightweight AIM-120 AMRAAM. Finally, an example of a long-range missile is the ducted rocket Meteor.

Three examples are shown in the bottom of Fig. 1.6 of the state-of-the-art drivers for air-to-surface missile types. The selected drivers are air-to-surface versatility, speed, and modularity. The examples are a short-range air-to-surface missile represented by the versatile AGM-114 Hellfire, a medium-range missile represented by the high-speed radar defense suppression AGM-88



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Fig. 1.7 Examples of surface-launched missile missions/types.

HARM, and a long-range missile represented by the precision strike modular Apache/Storm Shadow/Scalp.

Examples are shown in the top of Fig. 1.7 of tactical surface-to-surface missiles that illustrate the state-of-the-art drivers. The selected drivers are surface-to-surface size, modularity, and range. The example missiles are the man-portable Javelin anti-tank missile, the medium-range modular payload (e.g., dumb submunitions, smart submunitions, unitary surface warhead, penetrator warhead) MGM-140 ATACMS, and the long-range BGM-109 Tomahawk.

Surface-to-air missile state-of-the art drivers are shown in the bottom of Fig. 1.7. The selected drivers are surface-to-air weight, accuracy, and altitude. The missiles shown are the short-range lightweight FIM-92 Stinger, medium-range hit-to-kill PAC-3, and the long-range high-altitude SM-3 Standard Missile.

1.5 Aerodynamic Configuration Sizing Parameters

Aerodynamic configuration sizing parameters have an impact on missile requirements. Figure 1.8 is an assessment of the relative impact of aerodynamic configuration sizing parameters on flight performance measures of merit (e.g., weight, range, maneuver footprint, time to target). Also shown is the impact of aerodynamic configuration sizing parameters on other measures of merit (e.g., robustness, lethality, miss distance, observables, survivability, cost), as well as their impact on constraints (e.g., launch platform integration weight, length, and span constraints).

The aerodynamic configuration sizing parameters that are addressed in this text are listed on the left side of the figure. These are nose fineness, diameter, length, wing geometry/size, stabilizer geometry/size, flight control geometry/size, propellant/fuel, thrust profile, and flight conditions (α , M, h).

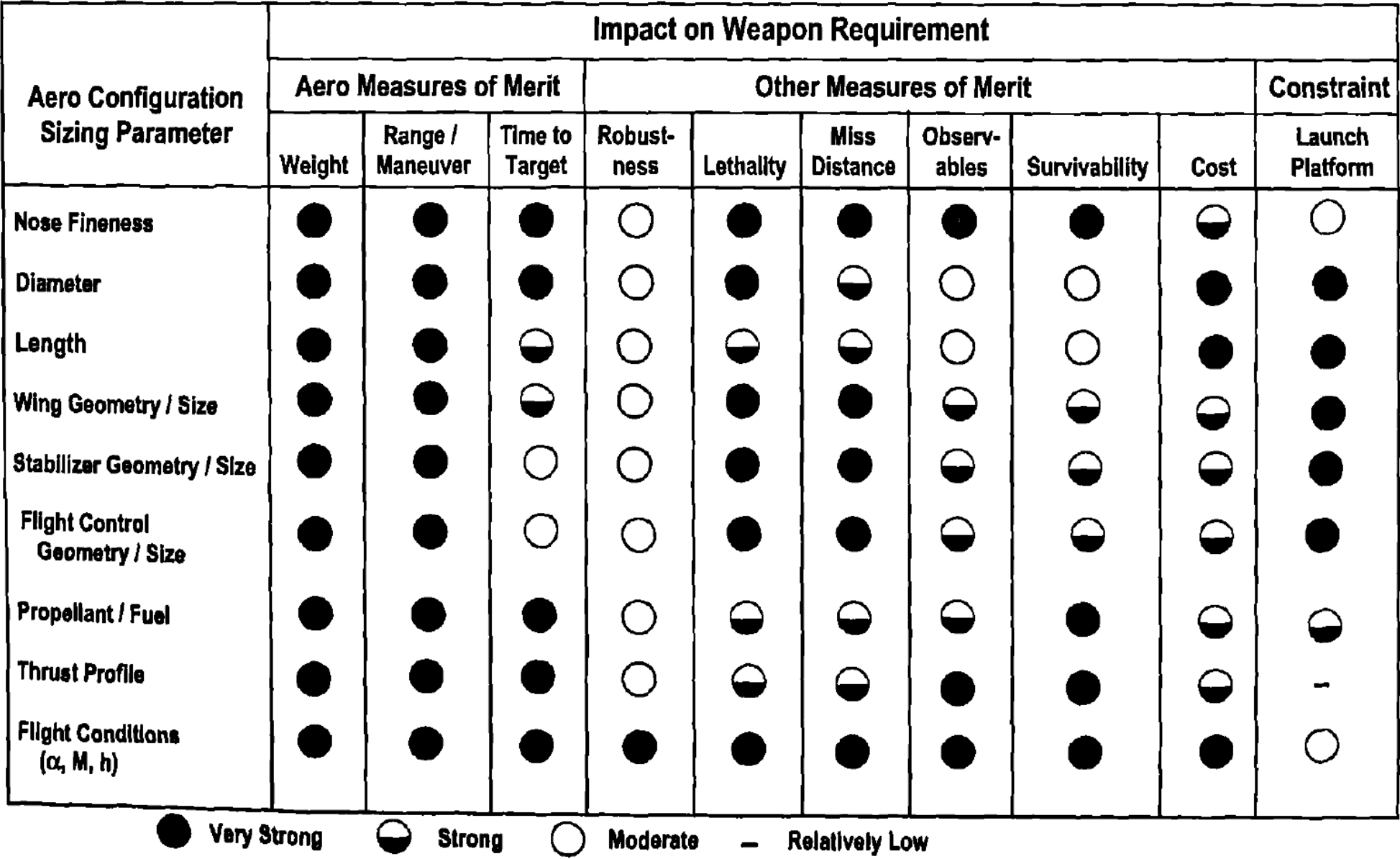


Fig. 1.8 Aerodynamic configuration sizing has high impact on mission requirements.

geometry/size, propellant/fuel, thrust profile, and flight conditions (angle of attack, Mach number, altitude). Note from the figure that the parameters have very strong impact on missile requirements in the areas of weight, range, and maneuverability. The aerodynamic configuration sizing parameters also have a strong impact in the areas of lethality, miss distance, and cost.

1.6 Example of Alternatives in Establishing Mission Requirements

Figure 1.9 is an example of alternative approaches that should be considered in establishing future mission requirements. It illustrates an assessment of alternative approaches for precision strike. The assessment includes the approaches that are used today by current systems, as well as a projection of capabilities that may be used in the future for new missile systems. Three measures of merit are assumed in comparing future precision strike missiles with the current systems. These are cost per shot, number of launch platforms required, and the effectiveness against time-critical targets (TCTs). For the current systems, two approaches are used: 1) penetrating aircraft with relatively short range subsonic precision guided munitions or missiles and 2) standoff platforms (ships, submarines, or aircraft) using subsonic cruise missiles. The penetrating aircraft systems include the F-117 with subsonic precision guided munitions such as JDAM. As shown, penetrating aircraft/subsonic precision guided munitions have an advantage of low cost per shot, about \$10,000 to \$20,000. However, the experience in Desert Storm showed that subsonic penetrating

Alternatives for Precision Strike	Measures of Merit		
	Cost per Shot	Number of Launch Platforms Required	TCT Effectiveness
<u>Future Systems</u>			
◆ Standoff platforms / hypersonic missiles	○	●	◐
◆ Overhead loitering UCAVs / hypersonic missiles	◐	◐	●
◆ Overhead loitering UCAVs / lightweight PGMs	●	○	◐
<u>Current Systems</u>			
◆ Penetrating aircraft / subsonic PGMs	●	—	—
◆ Standoff platforms / subsonic missiles	○	●	—
Note: ● Superior ◐ Good ○ Average — poor			

Note: C4ISR targeting SOTA for year 2010 projected to provide sensor-to-shooter/weapon connectivity time of less than 2 m and TLE of less than 1 m for motion-suspended target.

Fig. 1.9 Example of assessment of alternatives to establish future mission requirements.

aircraft do not have the capability to counter TCTs such as theater ballistic missiles (TBMs). Another current approach using standoff platforms such as ships, submarines, and large aircraft outside the threat borders requires fewer launch platforms, resulting in lower logistics costs. However, standoff platforms with subsonic cruise missiles (e.g., Tomahawk, CALCM) are also ineffective against TCTs.

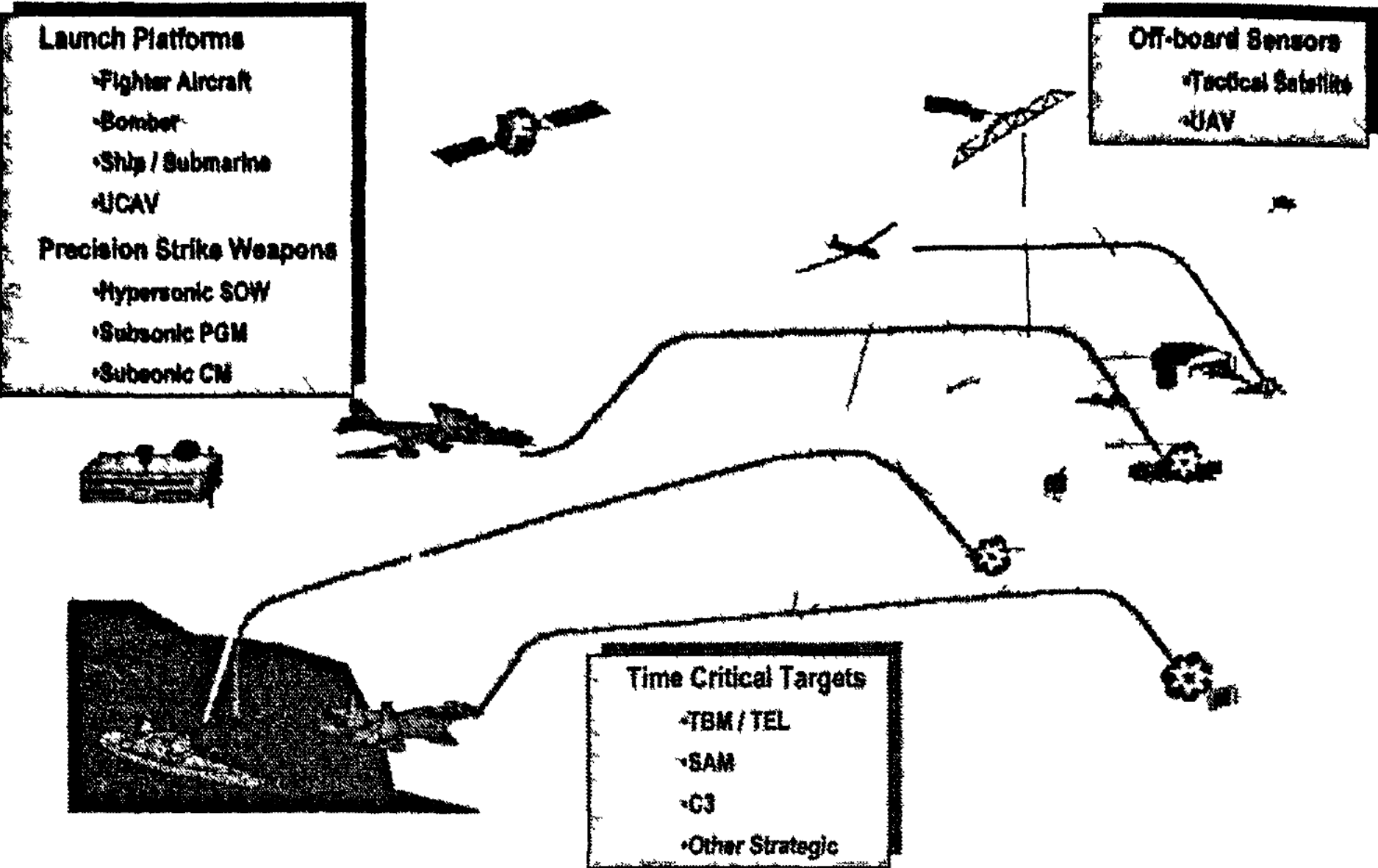
Also shown in the figure are future missile system alternatives for precision strike. Technology development work is under way in all three of the areas; the best approach has yet to be demonstrated. One approach is based on a standoff platform, with an aircraft, ship, or submarine standing off outside the threat country border. Hypersonic long-range precision strike missiles would provide broad coverage, holding a large portion of the threat country at risk. This approach is attractive because only a small number of launch platforms are required and there is good effectiveness against TCTs. Based on current technology programs, the cost of future hypersonic missiles is projected to be comparable to that of current cruise missiles.

Another potential alternative approach is to use overhead loitering unmanned combat air vehicles (UCAVs) with small hypersonic missiles. The number of UCAVs required is dependent upon the speed and range of their onboard missiles and the loiter altitude. A high loiter altitude tends to reduce the available payload weight/firepower. This approach would probably provide the fastest response time against TCTs because of the shorter required flight range of the missile.

A third approach is overhead loitering UCAVs with advanced lightweight precision guided munitions. Again, a design consideration is the loiter altitude required for survivability versus the weapon payload. This approach would have the lowest cost per shot but would also require a larger number of UCAVs.

An enabling synergistic capability for precision strike is the application of near-real-time, accurate targeting from either overhead tactical satellite or overhead unmanned air vehicle (UAV) sensors. It is projected that an advanced command, control, communication, computers, intelligence, surveillance, reconnaissance (C4ISR) network will be available around the year 2010 to support near-real-time and high accuracy targeting of TCTs. The C4ISR network could be used by all types of launch platforms (e.g., fighter aircraft, bombers, helicopters, UCAVs, ships, submarines, and ground vehicles). Illustrated in Fig. 1.10 are examples of a ground station, overhead satellite sensors and satellite relays, and the overhead UAV sensor platform elements of the assumed C4ISR architecture. The assumed C4ISR of the year 2010 is projected to have a capability for a target location error (TLE) of less than 1 m (1σ) for a motion suspended target and a sensor-to-shooter/weapon connectivity time of less than 2 min (1σ). A two-way data link from the launch platform to the missile will allow in-flight target updates and battle damage indication/battle damage assessment (BDI/BDA). An enabling capability for a lightweight/low-volume missile data link is phased array antenna technology. A phased array antenna can be conformally mounted on the missile airframe.

Figure 1.11 is an illustration of system-of-systems analysis. The example is a system analysis to develop mission requirements for precision strike in the year 2010 timeframe. A comparison is made of a subsonic cruise missile versus a



Note: C4ISR targeting state-of-the-art for year 2010 projected to provide sensor-to-shooter / weapon connectivity time of less than 2 m and target location error (TLE) of less than 1 m for motion suspended target.

Fig. 1.10 C4ISR tactical satellites and UAVs have high impact on mission capability. Figure also in color section.

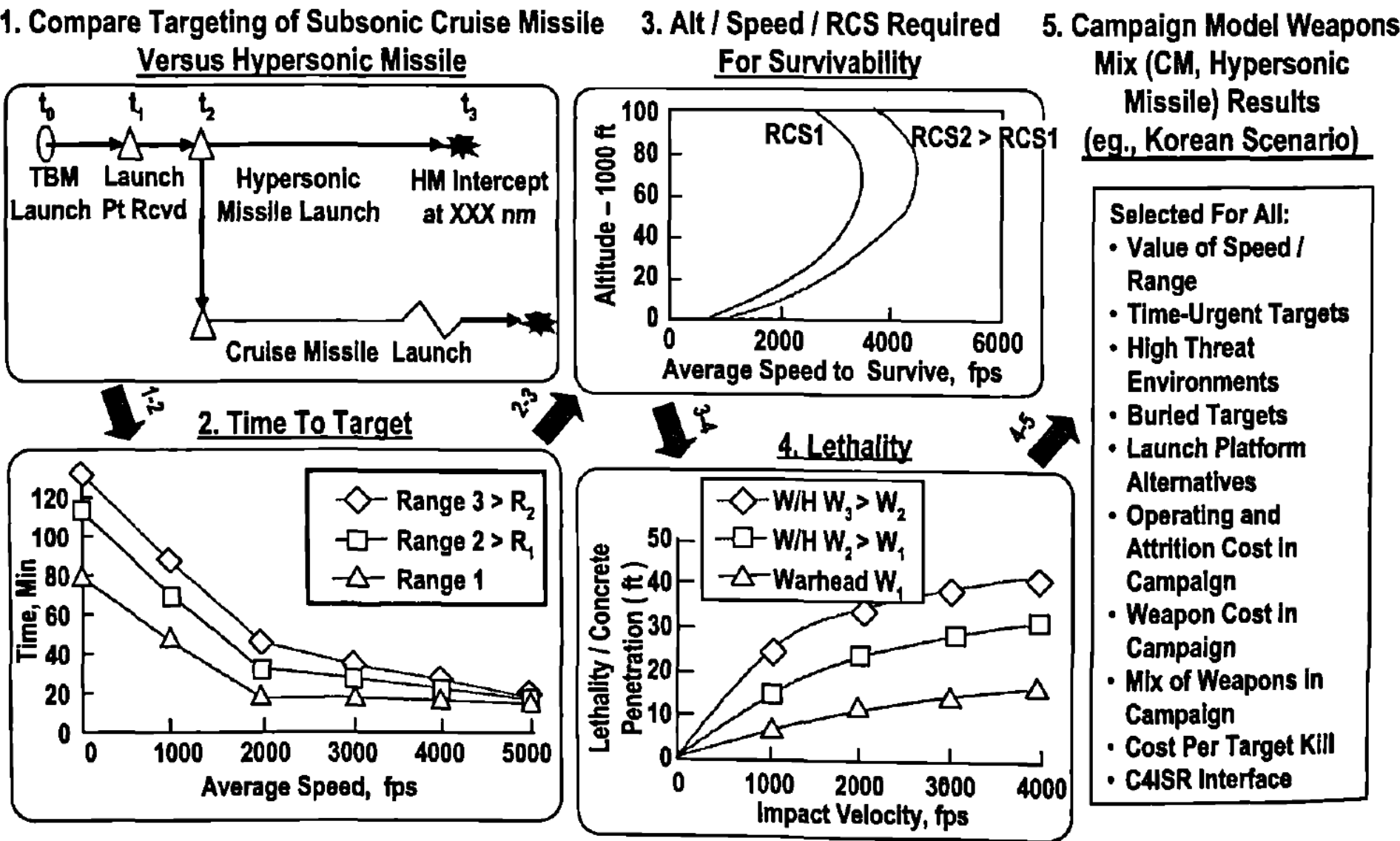


Fig. 1.11 Example of system-of-systems analysis to develop future mission requirements.

hypersonic missile. In the initial activity, requirements are defined for the targeting time of the two missiles. In the example both missiles rely on receiving the target location in near real time, with a fast response capability for launch. The second activity (shown in the lower left section of the figure) develops a solution matrix of time-to-target versus average speed. Shown in the figure are three assumptions of a relatively short-range R_1 for standoff against a relatively small country (e.g., North Korea), medium-range standoff R_2 , and long-range standoff R_3 against a relatively large country (e.g., Iran). Based on the knee of the curve of standoff range versus speed, an average speed is selected and a third activity is initiated, involving tradeoffs of flight attitude versus the level of reduced observables required for survivability. In the fourth activity (lower right section of the figure) a tradeoff is made of the missile lethality. In the example the impact velocity required for penetration of a buried target is evaluated as a function of warhead weight. Finally, a system-of-systems analysis campaign modeling activity is initiated (upper right section of the figure) to derive the weapon system-of-systems requirements such as the best mix of cruise missiles and hypersonic missiles for a representative scenario (e.g., Korean scenario). Measures of merit include the value of speed and range, capability against time-urgent targets, survivability in a high-threat environment, effectiveness against buried targets, launch platform alternatives (e.g., ship, aircraft, submarine), operating and attrition cost during the assumed campaign, cost of the number of weapons used during the campaign, mix of weapons (e.g., long-range hypersonic missile, long-range cruise missile, shorter range weapons), cost per target kill, and the requirements for the C4ISR interface.

Figure 1.12 is another illustration of the impact of mission requirements on missile design. It shows an example of technological surprise driving immediate mission requirements. The example is mission requirements for short-range air-to-air missiles. A comparison is made of the older technology AIM-9L Sidewinder with the technological surprise of the newer technology from the highly maneuverable Archer AA-11/R-73 missile. As a result of the technological surprise of the AA-11, there was a sudden mission requirement for the United States to develop the AIM-9X missile.

Shown on the left side of the figure are the off-boresight and flight range capability of the AIM-9L Sidewinder. The initial operational capability (IOC) for AIM-9L was the year 1977. The performance of the AIM-9L is ± 25 deg off boresight and approximately 6.5 n miles range. Contrasting with AIM-9L is a more recent missile, the Archer AA-11/R-73. The Archer is described in the right section of the figure. The Archer has a more recent IOC, 1987. Its performance is about ± 60 deg off boresight and 20 n miles range. New technologies developed for the Archer missile include thrust vector control, split canard, near-neutral static margin, ± 90 deg gimbaled seeker, and integration with a helmet-mounted sight. Archer also has forward vanes that weathercock, providing information on the missile angle of attack. In summary, Archer has higher maneuverability, greater off boresight, and longer range than AIM-9L.

Also shown at the bottom of the figure is a video illustrating the off-boresight advantage of a highly maneuverable missile such as Archer versus a relatively low maneuverable missile such as AIM-9L. In the video large off-boresight conditions and intensive pilot activities are required for short-range air-to-air

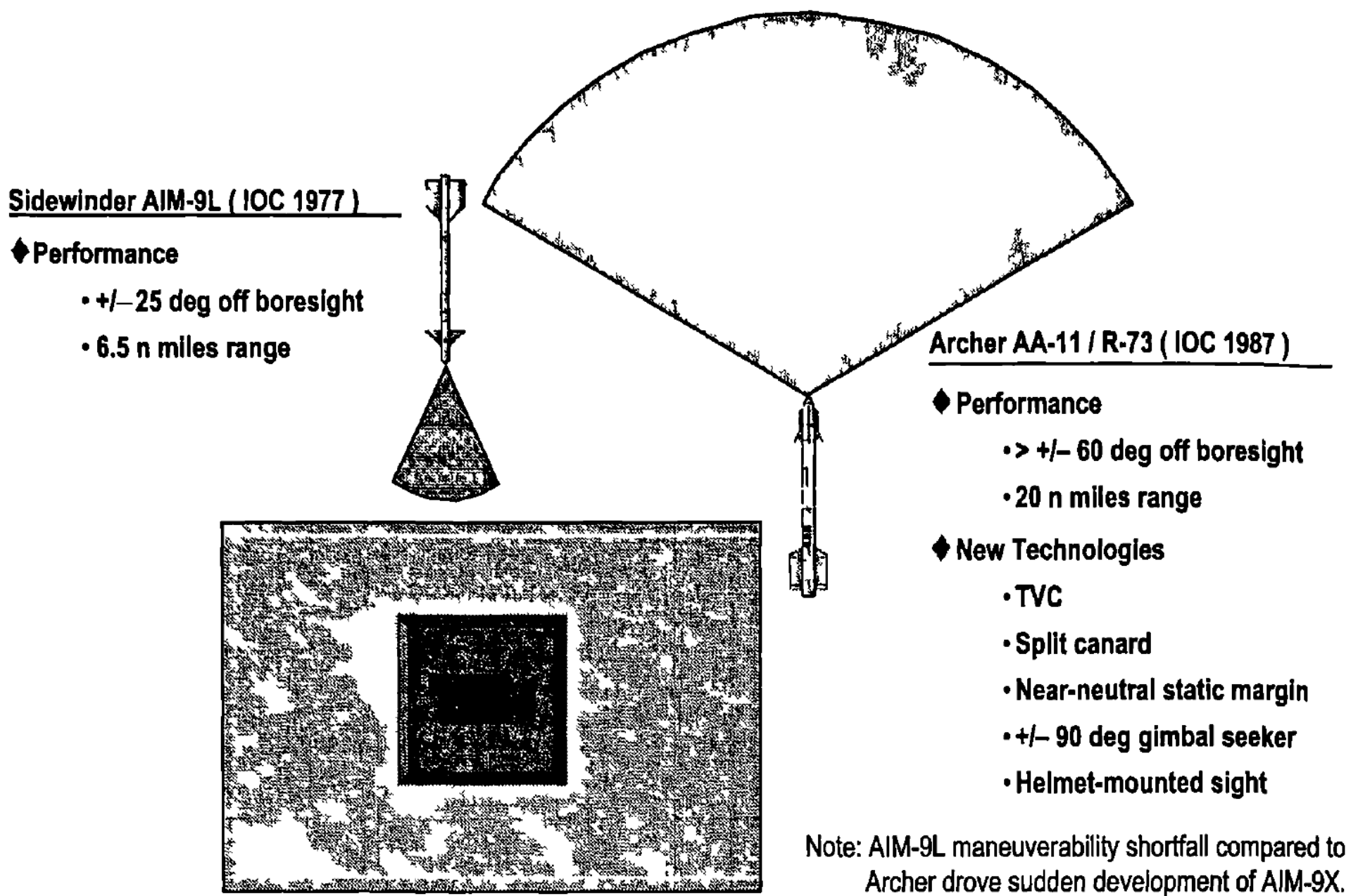


Fig. 1.12 Example of technological surprise driving immediate mission requirements. Video is in the CD included with this textbook.

combat. A highly maneuverable missile such as Archer can be launched at much larger off boresight than a low maneuverable missile such as AIM-9L, allowing an earlier launch in a short-range air-to-air engagement. Because an Archer-type missile has a superior short-range air-to-air capability, an older, relatively low maneuverability aircraft that uses a highly maneuverable, such as Archer, missile may be able to defeat a more modern, highly maneuverability aircraft that has an older, lower technology missile such as AIM-9L. The shortfall of AIM-9L maneuverability compared to the Archer was an immediate driver in the sudden development of AIM-9X. Like the Archer AA-11/R-73, AIM-9X has high maneuverability [through thrust vector control (TVC) and near-neutral static margin] and large off boresight (through a large gimbal angle seeker and integration with a helmet-mounted sight). In addition, AIM-9X has a high-resolution imaging infrared seeker for improved accuracy and counter-measures resistance.

1.7 Use of a Baseline Missile

Missile conceptual design synthesis is an evaluation of alternatives through an iterative process. Alternative configurations, alternative subsystems, and alternative technologies are evaluated in sizing the missile. As shown in Fig. 1.13, following the definition of mission requirements, a baseline missile is selected. For the purpose of this text, two alternative baselines are available for conceptual design, a rocket-powered medium-range missile based on the AIM-7 Sparrow

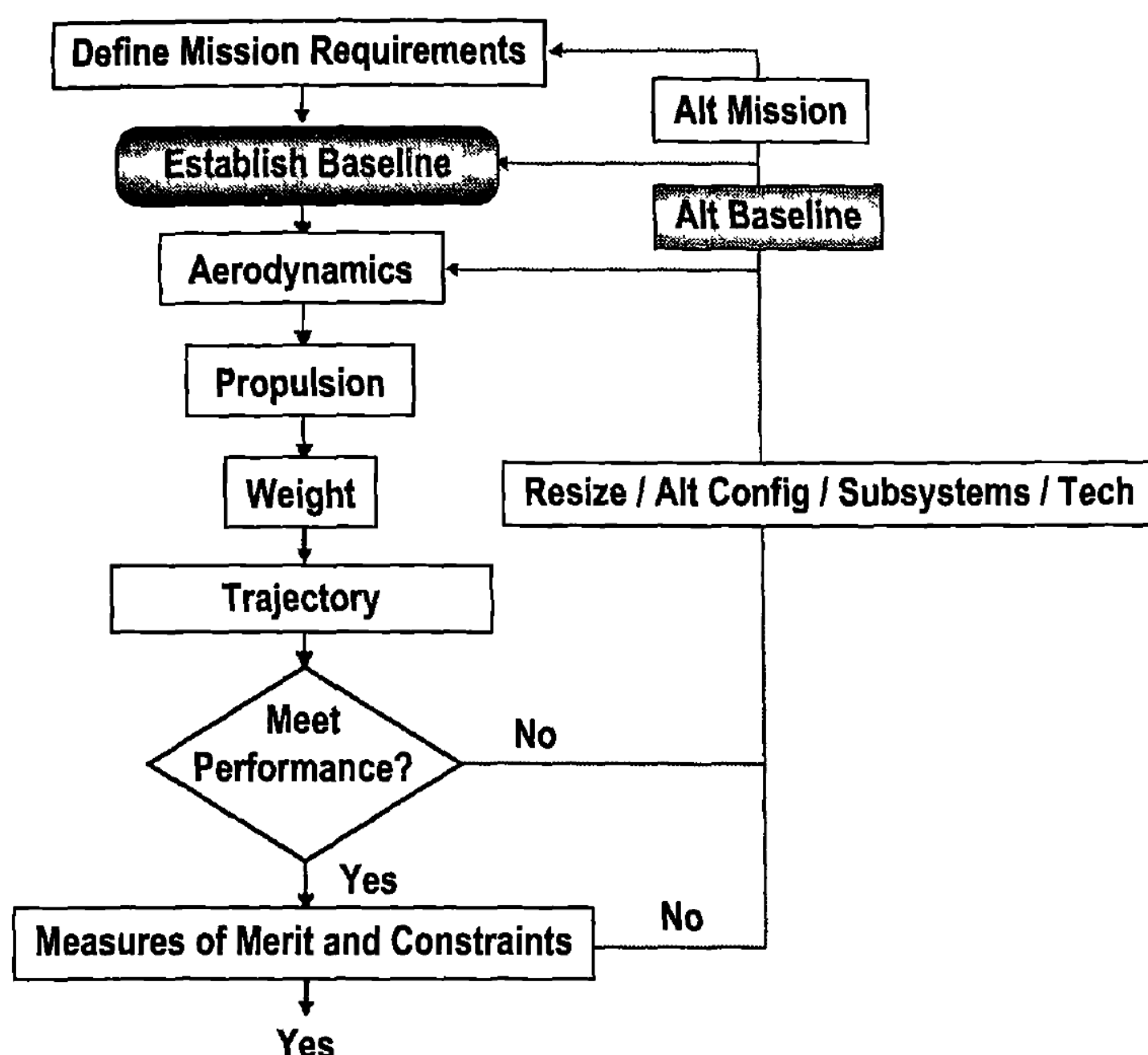


Fig. 1.13 Missile concept synthesis requires evaluation of alternatives and iteration.

and a ramjet-powered long-range missile based on the Advanced Strategic Air Launched Missile (ASALM). Detailed information on the baseline missiles is provided in Chapter 7.

Using a baseline missile to initiate the design process has benefits that include a faster design process and a more accurate design. Because a baseline missile is a real missile that is anchored with test data, it allows simple conceptual design methods to be used with reasonable accuracy. The existing documentation and the configuration control of a baseline missile provide traceability between cause and effect as the conceptual design process is conducted. System engineering has already been applied to the baseline missile, providing a benchmark for balanced subsystems. The startup time for conceptual design is greatly reduced by using a baseline missile. The default values for the conceptual design missile are the baseline missile. Sensitivity trends for the baseline missile indicate changes that would provide an improved design. However, note that one baseline cannot cover all of the range of possible solutions. The more data that are available for missile baselines, the better. Typically the propulsion system of the new conceptual missile should be the same type as that of the baseline missile. The baseline missile flight performance (e.g., range, Mach number, maneuverability) should also be comparable to that of the conceptual design missile. The baseline missile characteristics can usually be extrapolated about $\pm 50\%$ with reasonable accuracy.

Note that, when using a baseline missile, it is important not to get locked in by the baseline but to be creative in exploring design alternatives. Also, because the baseline is an existing missile, whereas the conceptual design usually involves a

missile projected for the future, the SOTA must often be projected into the future, particularly for the electronic and sensor subsystems.

Shown in Fig. 1.14 is an example of baseline missile data for conceptual design sizing. The example is based on the chin inlet, integral rocket ramjet baseline described in Chapter 7. In the upper left of the figure is an illustration of a configuration drawing of the baseline missile. The configuration drawing is a dimensioned layout with an inboard profile showing the major subsystems (guidance, warhead, fuel, booster/engine, and flight control surfaces). In the upper center of the figure are examples of tables for a missile weight statement and geometry data. Missile weight and center-of-gravity location are provided for launch, booster burnout, and engine burnout flight conditions. Weight and geometry data are also provided for the major subsystems. The upper right corner of the figure is an illustration of a description of the ramjet internal flow-path geometry. The internal flowpath geometry data include the inlet design capture area and the internal areas of the inlet throat, diffuser exit, flame holder plane, combustor exit, nozzle throat, and nozzle exit. Examples of aerodynamic data plots are illustrated in the left center section of the figure. The aerodynamic data of the ramjet baseline cover angles of attack up to 16 deg and Mach numbers up to 4.0. Aerodynamic coefficients and derivatives include zero-lift drag coefficient C_{D0} , normal force coefficient C_N , pitching moment coefficient C_m , pitching moment control effectiveness derivative $C_{m\delta}$, and normal force control effectiveness derivative $C_{N\delta}$. Examples of ramjet propulsion thrust T and the ramjet specific impulse I_{SP} are shown in the center of the figure. Thrust of the ramjet baseline is a function of Mach number, altitude, and fuel-to-air ratio. Specific impulse of the ramjet baseline is primarily a

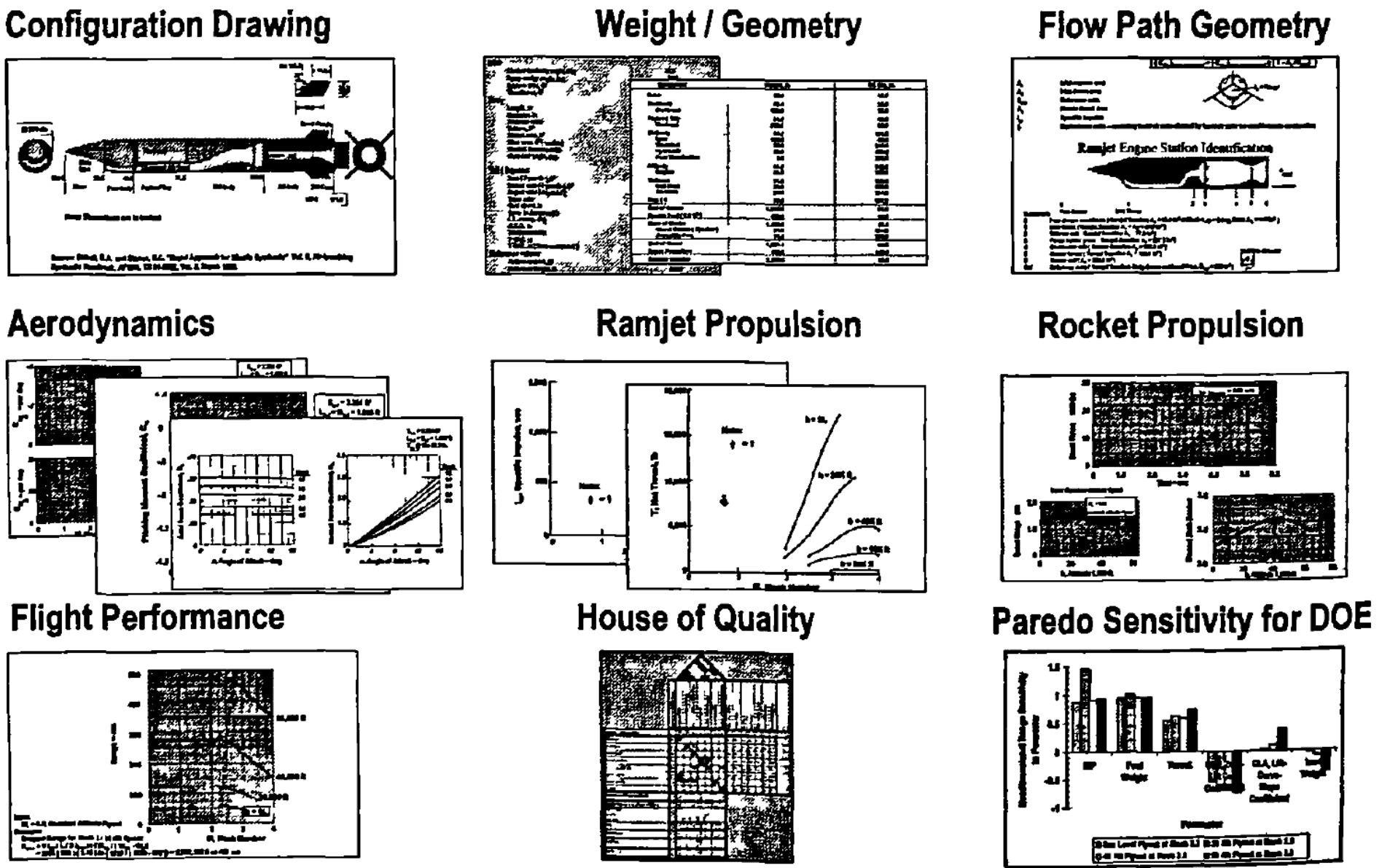


Fig. 1.14 Example of missile baseline data. Figure also in color section.

function of Mach number for fuel-to-air ratios less than stoichiometric. Rocket booster propulsion thrust, boost range, and burnout Mach number are illustrated in the right center of the figure as a function of launch Mach number and altitude. The left bottom section of the figure shows the maximum flight range of the ramjet baseline. Maximum flight range is a function of the launch Mach number, launch altitude, cruise Mach number, and the cruise altitude. In the left bottom center section is an example of a house of quality which includes a most important requirements weighting. The house of quality is developed through a consensus with the customer on the most important requirements and their relative weighting. It is used to map the “voice of the customer” to the “voice of the engineer.” The weighting from the house of quality is a tool for the downselect of the preferred concept alternative. Finally, the right bottom section of the figure is an example of a Pareto (ranking) sensitivity of design parameters to a measure of merit (e.g., maximum flight range). The Pareto sensitivity study is used to define a DOE set of alternative concepts for the design study. The sensitivity study in the example was conducted for cruise flight conditions ranging from Mach 2.3, sea level to Mach 3.0, 60,000 ft in altitude.

Shown in Fig. 1.15 is the process of using baseline missile data to provide enhanced accuracy to prediction methods. The predicted values of the design parameters from the conceptual design prediction methods are corrected by the baseline missile data. First, an uncorrected prediction is made of a notional conceptual design parameter $P_{CD,U}$. Second, an uncorrected prediction is made of the baseline missile notional parameter $P_{B,U}$. The correction in the prediction for the baseline missile $P_{B,C}/P_{B,U}$ provides a correction to the

$$P_{CD,C} = (P_{B,C} / P_{B,U}) P_{CD,U}$$

- ◆ $P_{CD,C}$ Parameter of conceptual design, corrected
- ◆ $P_{B,C}$ Parameter of baseline, corrected (actual data)
- ◆ $P_{B,U}$ Parameter of baseline, uncorrected (computed)
- ◆ $P_{CD,U}$ Parameter of conceptual design, uncorrected (computed)
- ◆ **Example** 
 - Ramjet Baseline with RJ-5 fuel (heating value = 11,300,000 ft-lbf / lbm)
 - Advanced Concept with slurry fuel (40% JP-10 / 60% boron carbide = 18,500,000 ft-lbf/lbm)
 - Flight conditions: Mach 3.5 cruise, 60k ft altitude, combustion temperature 4,000 R
 - Calculate specific Impulse $(I_{SP})_{CD,C}$ for conceptual design, based on corrected baseline data
 - $(I_{SP})_{B,C} = 1,120$ s
 - $(I_{SP})_{B,U} = 1387$ s
 - $(I_{SP})_{CD,U} = 2,271$ s
 - $(I_{SP})_{CD,C} = [(I_{SP})_{B,C} / (I_{SP})_{B,U}] (I_{SP})_{CD,U} = [(1120) / (1387)] (2271) = 0.807 (2271) = 1,834$ s

Fig. 1.15 Baseline design data allows the correction of computed parameters in conceptual design.

prediction method. Finally, the corrected conceptual design parameter $P_{CD,C}$ is computed by

$$P_{CD,C} = (P_{B,C}/P_{B,U})P_{CD,U} \quad (1.1)$$

In the example the ramjet baseline fuel is RJ-5, with a heating value of 11,300,000 ft-lbf/lbm (14,525 BTU/lbm). The assumed fuel of the more advanced conceptual design missile is a high-density slurry fuel consisting of 40% JP-10 and 60% boron carbide. It has a higher heating value of 18,500,000 ft-lbf/lbm. The assumed cruise condition of this example is Mach 3.5, 60,000 ft in altitude. The ramjet combustion temperature is assumed to be 4,000°R. Finally, the uncorrected specific impulse predicted for the high-density fuel $[(I_{SP})_{CD,U} = 2,271 \text{ s}]$ is corrected by the ratio of the assumed actual specific impulse of the baseline $[(I_{SP})_{B,C} = 1,120 \text{ s}]$ to the predicted specific impulse of the baseline $[(I_{SP})_{B,U} = 1,387 \text{ s}]$, giving a factor of 0.807 correction to the specific impulse for the high-density fuel. The corrected specific impulse is

$$\begin{aligned} (I_{SP})_{CD,C} &= [(I_{SP})_{B,C}/(I_{SP})_{B,U}](I_{SP})_{CD,U} \\ &= (1120/1387)(2271) = 0.807(2271) = 1,834 \text{ s} \end{aligned} \quad (1.2)$$

1.8 Summary

This chapter provided an overview of the missile design process. Also included were examples of tactical missile characteristics. A comparison was made of tactical missile capabilities with those of combat aircraft. In many cases tactical missiles drive the SOTA more than combat aircraft. The aerodynamic configuration sizing parameters of tactical missiles were identified. It was shown the aerodynamic configuration sizing parameters have a strong impact on the missile system measures of merit and constraints. The chapter also gave examples of the process and considerations for establishing mission requirements. Included was an example of defining the mission requirements for a hypersonic strike missile based on an analysis of a mix of hypersonic strike missiles with subsonic cruise missiles. Another example was the sudden development of a mission requirement for a highly maneuverable air-to-air missile based on a technology surprise that pointed out the shortfall in current capability. Finally, Chapter 1 presented a process for correcting design predictions using baseline missile data. The following chapters of this text will address the specific methods and technologies for tactical missile design.

Problems

- 1.1 The missile design team should address the areas of mission/scenario definition, weapon requirements, launch platform integration, design, and t_____ a _____.

- 1.2 The steps to evaluate missile flight performance require computing aerodynamics, propulsion, weight, and flight t_____.
- 1.3 Air-to-air missile characteristics include light weight, high speed, and high m_____.
- 1.4 Air-to-surface missiles are often versatile and m_____.
- 1.5 Four aeromechanics measures of merit are weight, range, maneuverability, and t___ to target.
- 1.6 The launch platform often constrains the missile span, length, and w_____.
- 1.7 An enabling capability for hypersonic strike missiles is fast and accurate C_____.
- 1.8 An enabling capability for large off-boresight air-to-air missiles is a h_____ m_____ sight.
- 1.9 A baseline design improves the accuracy and s_____ of the design process.